

Similarity by Selected Traits

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can represent items with numeric features.

I can calculate a simple distance.

I can explain how units and chosen features affect similarity.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

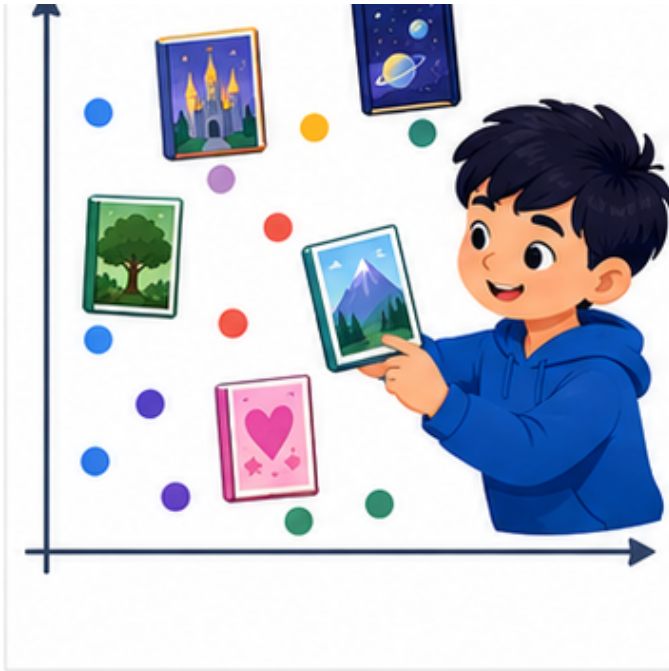
Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 21: Similarity by Selected Traits

Different feature choices produce different notions of closeness.



THINKER MISSION

Compute simple distance on genre, length, and difficulty.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

What does $\text{abs}(3 - 7)$ measure? If unfamiliar, calculate the positive gap.

This week's end goal

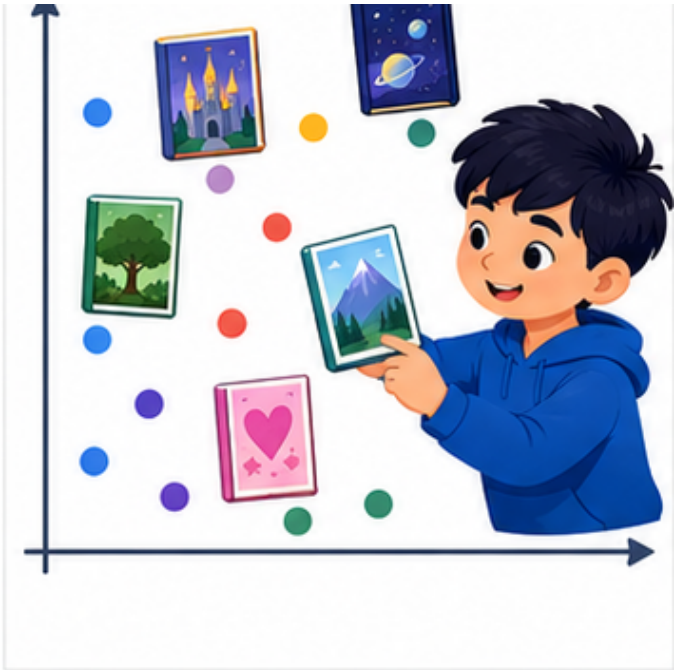
By the end, I can calculate a feature distance and state how scaling changes its meaning.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use similarity by selected traits to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

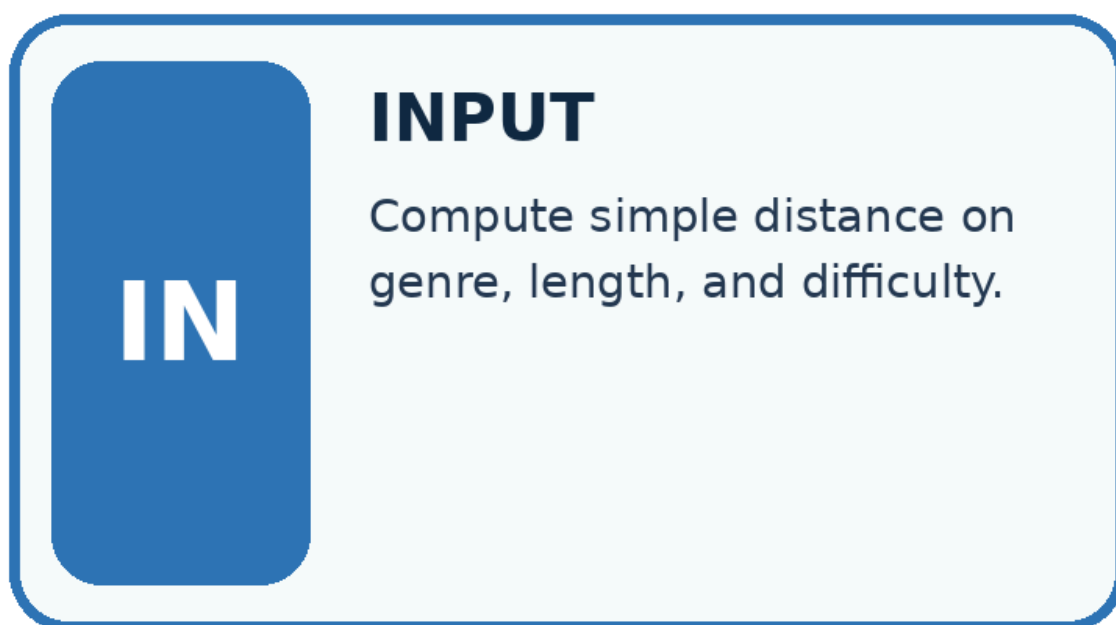
A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 scale. A learner requests [2, 4]. We compare options by adding absolute differences. This is a chosen rule for similarity; it is not a learned model or an objective measure of quality.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

Different feature choices produce different notions of closeness.



OUT

OUTPUT

Compute simple distance on genre, length, and difficulty.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
target = [2, 4]

options = {"Sketch": [1, 3], "Relay": [5, 4]}

for name in options:

    features = options[name]

    distance = (abs(features[0] - target[0])

                + abs(features[1] - target[1]))

    print(name, distance)
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Sketch [1, 3]

Evidence: 1 + 1

Meaning: Distance 2

Step or case: Relay [5, 4]

Evidence: 3 + 0

Meaning: Distance 3

Read each row from left to right and connect the evidence to the stated rule.

abs gives a nonnegative gap. Under this rule Sketch is closer. Feature order must match: first energy, then group work. Both scales are 0 to 5, but that does not prove one unit has identical practical meaning in both features.

Calculate a gap one feature at a time

Keep the features in the same order: energy first, group work second. For target [2,4] and Sketch [1,3], the gaps are $\text{abs}(1-2)=1$ and $\text{abs}(3-4)=1$; the distance is 2. Relay [5,4] has gaps 3 and 0, so its distance is 3. Smaller means closer under this particular rule. It does not mean better in every way, because the two recorded features leave out other needs.

Connect the picture and the explanation

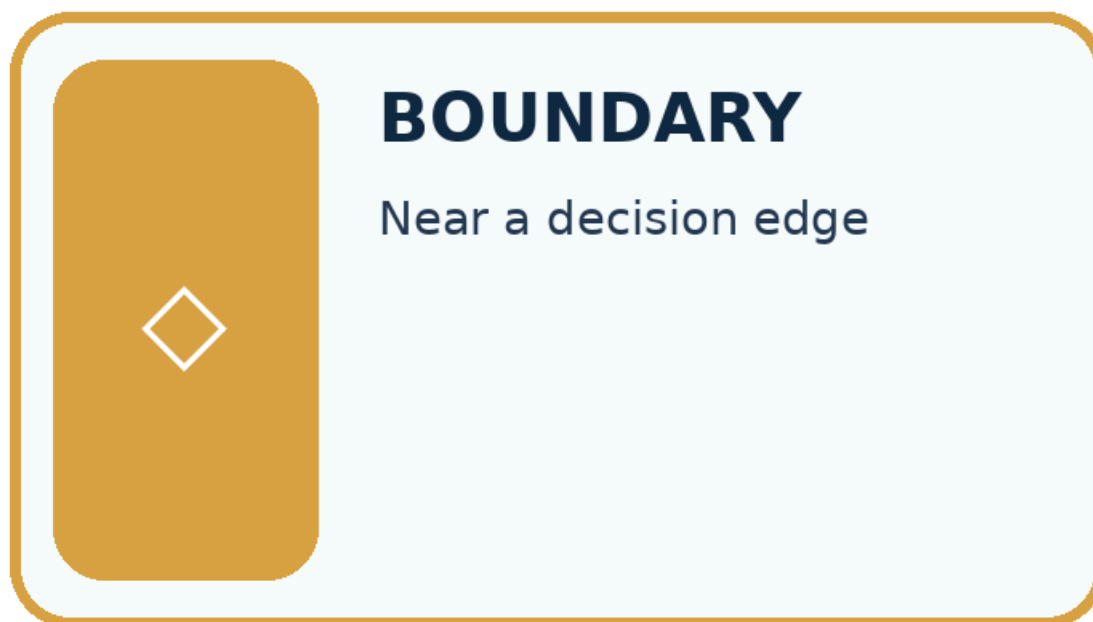
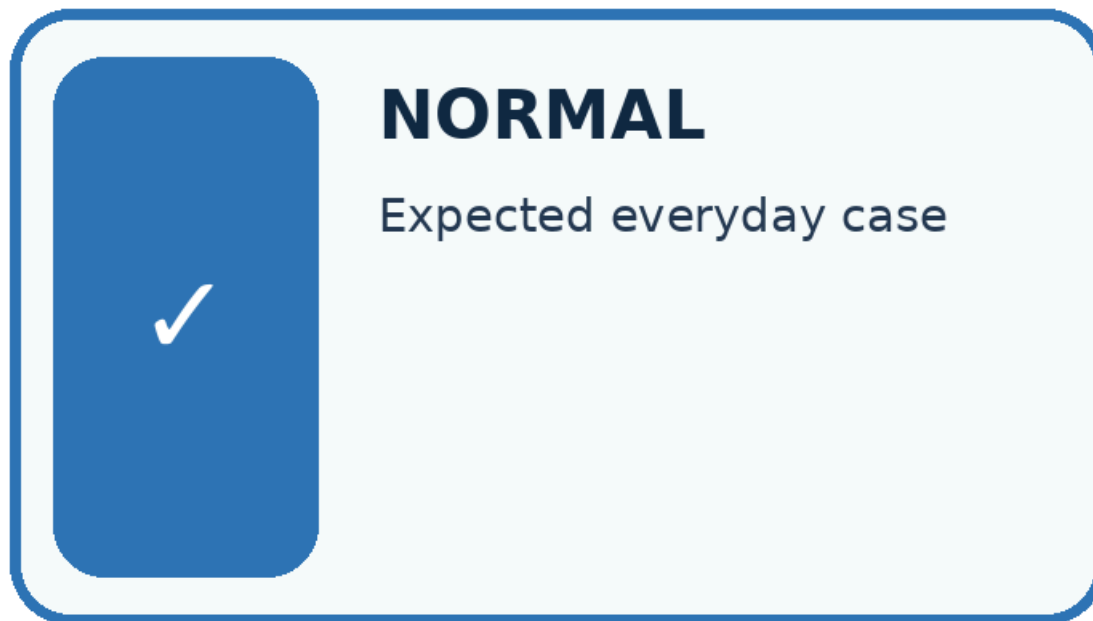
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Give the first-coordinate difference weight 2 while the second keeps weight 1.

Prediction: Distances become $A=3$, $B=4$, $C=3$; A and C tie.

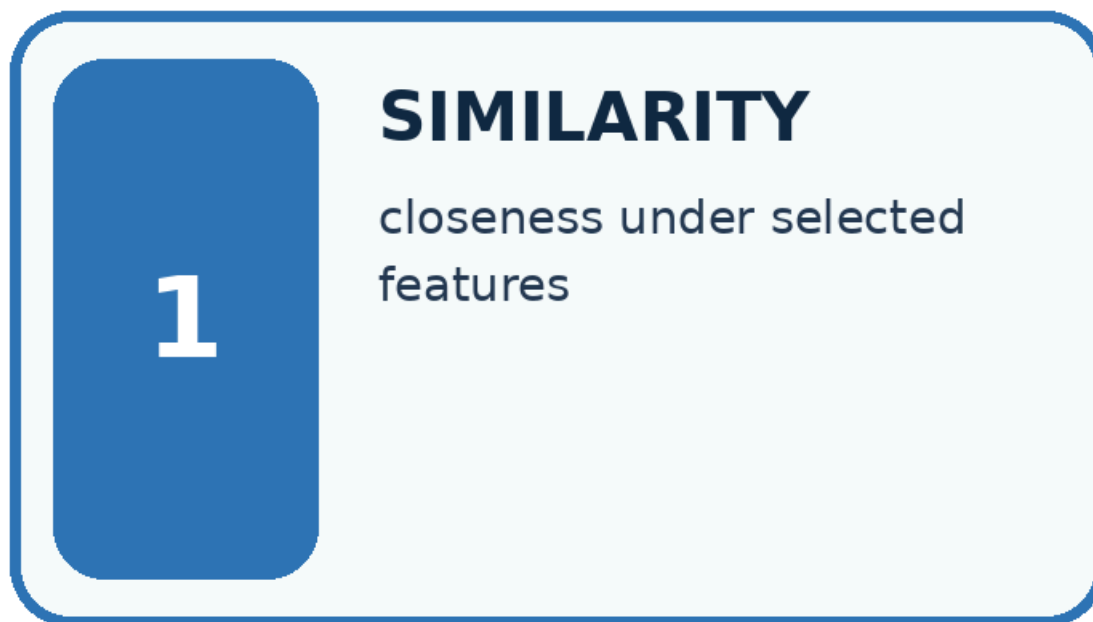
Reason: The representation's comparison rule changes; the supplied vectors stay the same.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

DISTANCE

number measuring
difference

3

COORDINATE

feature value used as a
position

4

SCALE

how feature values are measured

Word · My own example · Not this / boundary

Word: similarity

My own example: How alike two items are according to selected features and a declared comparison method.

Not this / boundary: _____

Word: distance

My own example: A specified measure of how far apart two representations are.

Not this / boundary: _____

Word: coordinate

My own example: A value identifying position along an agreed axis or within a space.

Not this / boundary: _____

Word: scale

My own example: The units and range used to represent or compare a quantity.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Feature vector

Meaning: An ordered list of numeric features

Word: Manhattan distance

Meaning: The sum of absolute feature differences

Word: Normalization

Meaning: Rescaling features to a comparable scale

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 21 • WORKED EVIDENCE

- 1 Reader target: length=2, difficulty=3.
- 2 Book A: (2,2) → grid distance $|2-2|+|2-3|=1$.
- 3 Book B: (4,3) → distance $|4-2|+|3-3|=2$.
- 4 A is closer under these features; changing features can change the answer.

1. Reader target: length=2, difficulty=3.
2. Book A: (2,2) → grid distance $|2-2|+|2-3|=1$.
3. Book B: (4,3) → distance $|4-2|+|3-3|=2$.

4. A is closer under these features; changing features can change the answer.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Find the distance from target [2, 4] to [2, 4] and to [3, 3].

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Similarity by Selected Traits

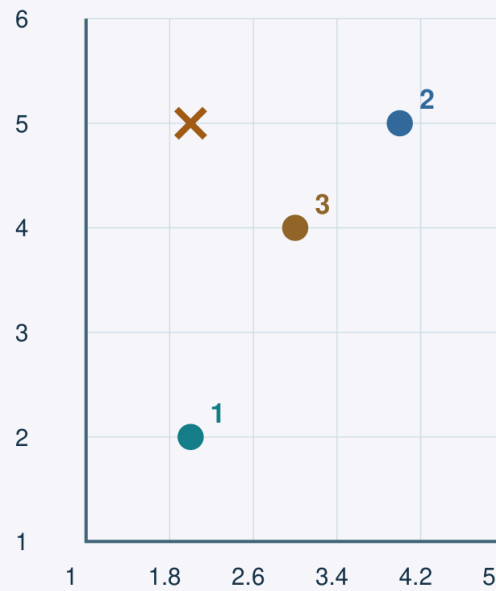
A fictional toy book matcher represents two preferences with numeric features.

The facts we will use

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use Manhattan distance: add absolute coordinate differences; return all smallest-distance candidates.

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Distance A

$$|2-2|+|5-2|=0+3=3.$$



2. Distance B

$$|2-4|+|5-5|=2+0=2.$$



3. Distance C

$$|2-3|+|5-4|=1+1=2.$$



4. Choose closest

B and C tie at distance 2; keep the tie visible.

CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Distance A

What happens and why: $|2-2|+|5-2|=0+3=3.$

Step: 2. Distance B

What happens and why: $|2-4|+|5-5|=2+0=2.$

Step: 3. Distance C

What happens and why: $|2-3|+|5-4|=1+1=2$.

Step: 4. Choose closest

What happens and why: B and C tie at distance 2; keep the tie visible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Each coordinate contributes according to the declared distance rule. Smaller distance means closer in this representation, not automatically a better reading experience. If one coordinate uses a much larger numerical scale, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

What this example does not establish

A toy numeric encoding does not establish meaningful interpersonal distances or suitability of a recommendation.

Read the labelled picture: Explain why B ceases to be a closest option without any candidate data changing.

Change one thing: Give the first-coordinate difference weight 2 while the second keeps weight 1. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Similarity is an objective fact independent of chosen features. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Similarity depends on the features collected and the distance rule. Missing accessibility, cost or preference information can make the closest item unsuitable. Hard requirements should be checked separately.

Two activities have distance zero. Must they suit every learner equally well?

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

If one feature is duration in minutes up to 60 and another a rating up to 5, raw distance may be dominated by duration. Specify scaling or weights before ranking; do not silently change units to get a preferred winner.

Practise with less help: Compute distances

Use workbook W2 for the connected example “Distance between feature vectors”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Distance between feature vectors. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Explain why B ceases to be a closest option without any candidate data changing.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Plot six books on a 2-feature grid, compute simple grid distance, find nearest, then change one selected feature and compare.

Plot six books on a 2-feature grid, compute simple grid distance, find nearest, then change one selected feature and compare.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
 - 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
 - 3 • APPLY: Solve a different example independently.
 - 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
 - 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Make selected traits and their scales visible in every recommendation explanation.

THIS WEEK'S CHECKPOINT Make selected traits and their scales visible in every recommendation explanation.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Compute simple distance on genre, length, and difficulty.

CHALLENGE Compute simple distance on genre, length, and difficulty.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Traits are [animals,adventure]. $A=[1,1]$, $B=[1,0]$, $C=[0,0]$; query= $[1,0]$.

The question

Use number of differing traits as distance.

Worked explanation

Distances: $A=1$, $B=0$, $C=1$. B is nearest with equal feature weights.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Count differing traits from query [1,0]

Evidence or stage · Value or result

Evidence or stage: A [1,1]

Value or result: 1

Evidence or stage: B [1,0]

Value or result: 0

Evidence or stage: C [0,0]

Value or result: 1

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as distance.

Distances: A=1, B=0, C=1. B is nearest with equal feature weights.

14. Try a new case independently

Independent case: Rank independently

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate distances, select the closest, and identify a tie among the others.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Score similarity using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Make selected traits and their scales visible in every recommendation explanation.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional toy book matcher represents two preferences with numeric features.

By the end, I can · How I will show it

By the end, I can: Represent items with numeric features

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Calculate a simple distance

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Explain how units and chosen features affect similarity

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Represent items with numeric features

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy book matcher represents two preferences with numeric features.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B and C tie at distance 2; keep the tie visible.

1. Distance C
2. Choose closest
3. Distance A
4. Distance B

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Choose closest B and C tie at distance 2; keep the tie visible. Each coordinate contributes according to the declared distance rule. Smaller distance means closer in this representation, not automatically a better reading experience. If one coordinate uses a much larger numerical scale, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

Practice 2: explain the deciding evidence

Changed-case practice: Give the first-coordinate difference weight 2 while the second keeps weight 1. Which conclusion is supported by the supplied facts?

1. Distances become $A=3$, $B=4$, $C=3$; A and C tie.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must

say what follows from those facts.

After your attempt, compare with the supported explanation: Distances become $A=3$, $B=4$, $C=3$; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Repair a missing building block

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?. Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf predictions from training examples. Short practice: Choose the best original question and explain why its more balanced split matters here.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.